



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

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Course Name: Solar Energy Engineering
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Instructor: Assoc. Prof. Dr. Murat Özdenefe
Assistant: Hussain Shawish
Assessment: Project
Due Date: 7 June 2026 until 23:59 hrs.

PROJECT NAME: PV SYSTEM DESIGN

STUDENT(S) NO.: 23704019, 22900037

NAME(S), SURNAME(S): TARIQ TARIK, DANIEL ADEBAYO

SECTIONS	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract	2, 7	9	5		
Introduction and scope	2, 7	9	10		
Building model	2, 7	9	10		
Modelling and Simulation Parameters	2, 7	9	30		
Results and discussion	2, 7	9	25		
Conclusion	2, 7	9	10		
References & appendix	2, 7	9	10		
					Marks for report format are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Understand the characteristics of solar energy, sun-earth geometric relations and sun path/shade diagrams.	X						
2	Comprehend beam & diffuse rad., and estimation of terrestrial rad. intercepted & absorbed by surfaces.	X						
3	Have a fundamental knowledge regarding the heat transfer aspects of the solar thermal design.	X						
4	Be able to analyze flat plate & concentrating solar thermal collectors.	X						
5	Develop an understanding of thermal energy storage with emphasis placed on water.	X						
6	Analyze solar thermal energy component models such as collector heat exchangers and array of collectors.	X						
7	Understand and use f-chart design methods for designing typical liquid based solar heating systems.	X	X					
8	Get familiar with basics of PVs & calculate electrical capacity of a PVT system from various measurements.	X					X	
9	Be able to design and analyze various solar systems by employing contemporary simulation tools.		X					X
Weight of Student Outcomes		H	H				L	H

Instructions:

- A. Report presentation:** Write relevant equations and your assumptions. Draw neat, labeled diagrams where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations/standards used. References should be from authentic sources. Include the relevant standards (preferably in the Appendix).
- B. Submission:** Upload your report (as a pdf file) and supplementary files to the "Project and Lab Work" section in LMS within the designated time. The file uploaded should be named as [MENG442-Project-Student ID(s)-Student Name(s)]. No late submissions will be accommodated.

ABSTRACT

The design and simulation of a photovoltaic (PV) system for a workshop building in Famagusta, North Cyprus, are presented in this project. Using DesignBuilder software, the building was modeled while considering various operational and environmental factors, including lighting, HVAC systems, occupancy schedules, and local weather conditions. Through the integration of solar energy technology, an appropriate photovoltaic system was designed to meet the building's annual electricity demand. Suitable PV panels and inverter systems were selected based on system compatibility and performance requirements. DesignBuilder simulations showed that the proposed system can generate sufficient electricity to satisfy the building's energy needs while promoting sustainable and environmentally responsible energy use.

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1. INTRODUCTION AND SCOPE

The growing energy demand around the world for sustainable and eco-friendly energy systems has led to the increase in the adoption of renewable energy technologies, particularly photovoltaic (PV) systems. Today, solar energy is one of the most reliable and fastest growing renewable energy resources because it can produce electricity directly from sunlight without any harmful emissions. Photovoltaic systems are built on the photovoltaic effect of semiconductor materials that transform solar radiation into electricity. This technology has major advantages including low operating costs, reduced greenhouse gas emissions and long-term energy sustainability.

Famagusta, located in North Cyprus, possesses favorable climatic conditions for solar energy applications because of its high solar irradiance levels and extended sunshine duration throughout the year. These conditions make the region highly suitable for PV system integration into commercial and industrial buildings. In this project, a photovoltaic system is designed for a three-story workshop building located in Famagusta. The building consists of a basement, ground floor, and first floor, and the annual electricity demand of the structure was targeted to be approximately 150,000 kWh through detailed modeling and simulation processes.

The goal of this project is to design and assess a photovoltaic (PV) system that is capable of fulfilling the majority of the electrical load of a workshop building and improve energy sustainability and decrease grid dependency. The building was modelled in DesignBuilder software with a realistic occupancy schedule, HVAC operations, lighting loads, equipment usage and construction material properties to accurately simulate real operating conditions.

The scope of this project includes:

- Modeling the workshop building using DesignBuilder software.
- Defining occupancy schedules, lighting loads, equipment loads, and HVAC operation.

- Simulating annual electrical energy consumption.
- Determining the optimal PV system size and installation parameters.
- Evaluating the annual PV electricity generation potential using climatic data for Famagusta.
- Comparing the generated solar energy with the building's annual electricity demand.
- Assessing the feasibility of achieving improved energy efficiency and reduced dependence on grid electricity through PV integration.

2. BUILDING MODEL

The building used in this project is a three-story workshop facility in Famagusta, North Cyprus, consisting of a basement, ground floor and first floor. The AutoCAD drawings indicated that the building was oriented approximately 19° west of true north, a critical consideration for evaluating solar exposure and determining the optimal positioning of photovoltaic panels. First, the building geometry was created in AutoCAD and then converted to DXF format and imported into DesignBuilder software for detailed modeling and simulation.

The model included all walls, openings, thermal zones, and construction materials to ensure accurate energy analysis. Reinforced concrete and brick walls were selected according to the project requirements, while local weather data for Famagusta was integrated into the simulation to account for environmental conditions such as solar radiation and temperature. The roof area was selected for PV panel installation due to its favorable exposure to sunlight, allowing improved solar energy generation and system efficiency.

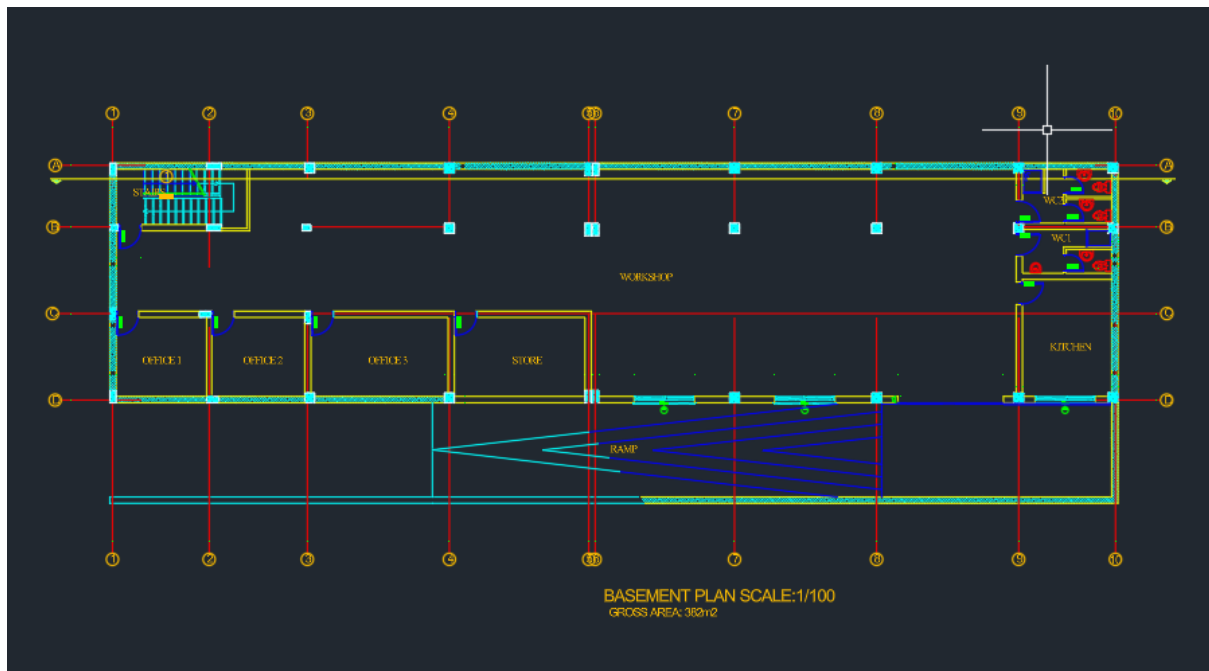


Figure 1; AutoCAD drawing of the basement plan

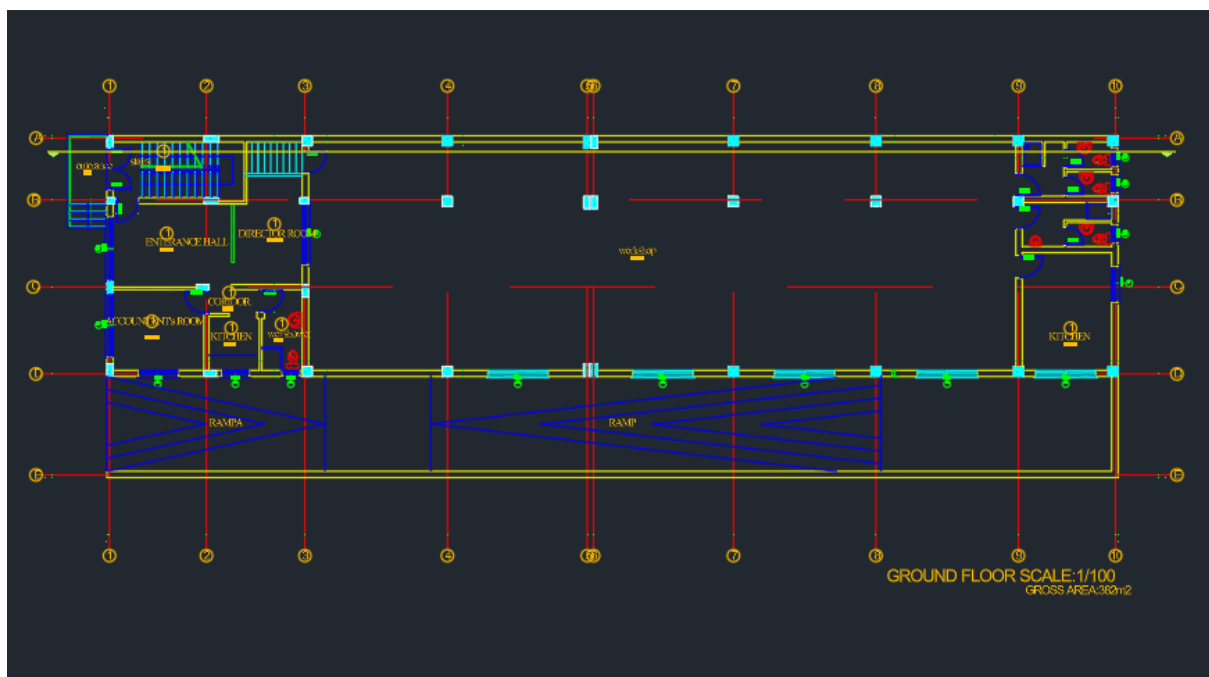


Figure 2; AutoCAD drawing of the ground floor plan

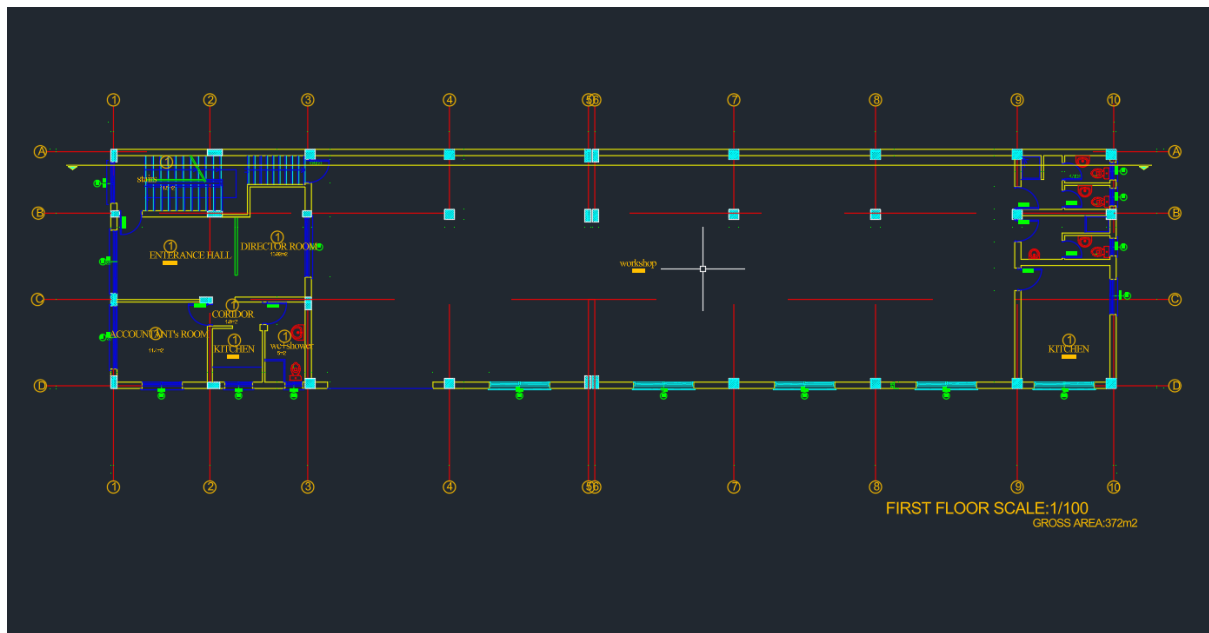


Figure 3; AutoCAD drawing of the first-floor plan

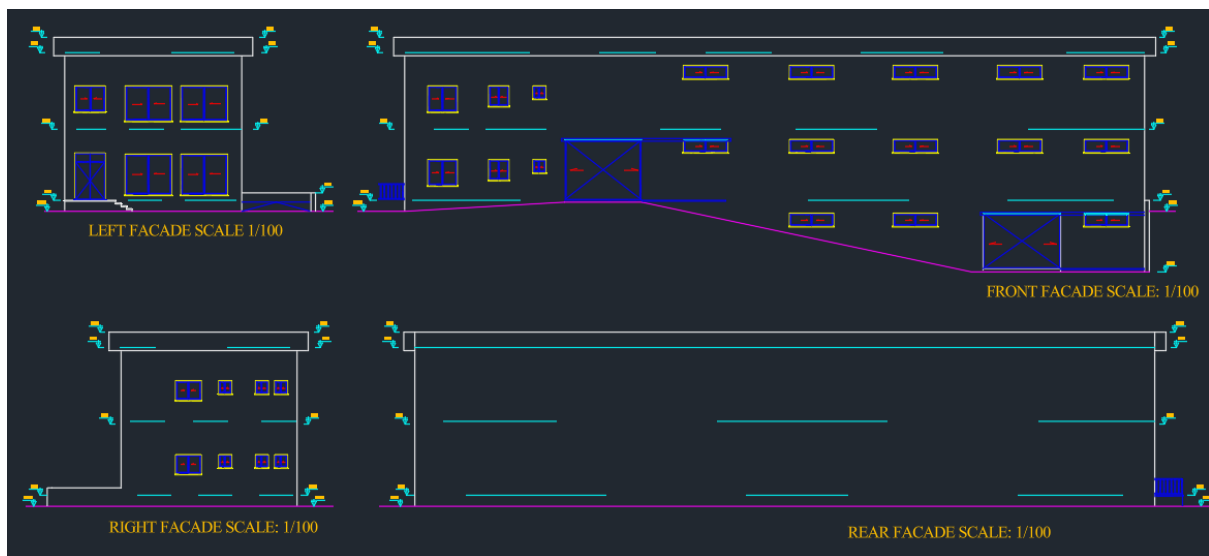


Figure 4; Various side views of the building on AutoCAD software

The AutoCAD file was then converted into a DXF format to enable its import into the DesignBuilder software. Using the available modeling tools, the building was successfully modeled, including the underground basement level. Special attention was given to accurately modeling all walls in contact with the ground, as indicated in the AutoCAD drawings. These

elements were assigned using the Component Block tool, with the block type set to “Ground,” as shown below.

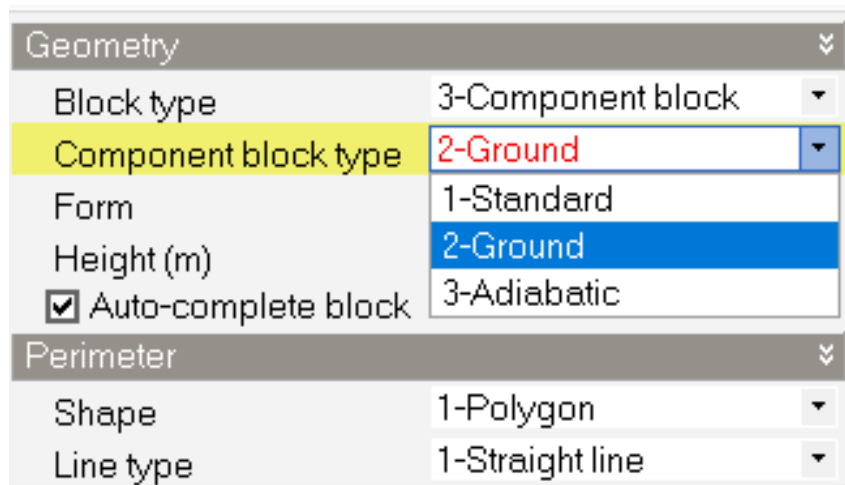


Figure 5; Assigning components on DesignBuilder

The weather data for Famagusta was obtained from <https://climate.onebuilding.org/> and loaded to DesignBuilder software.

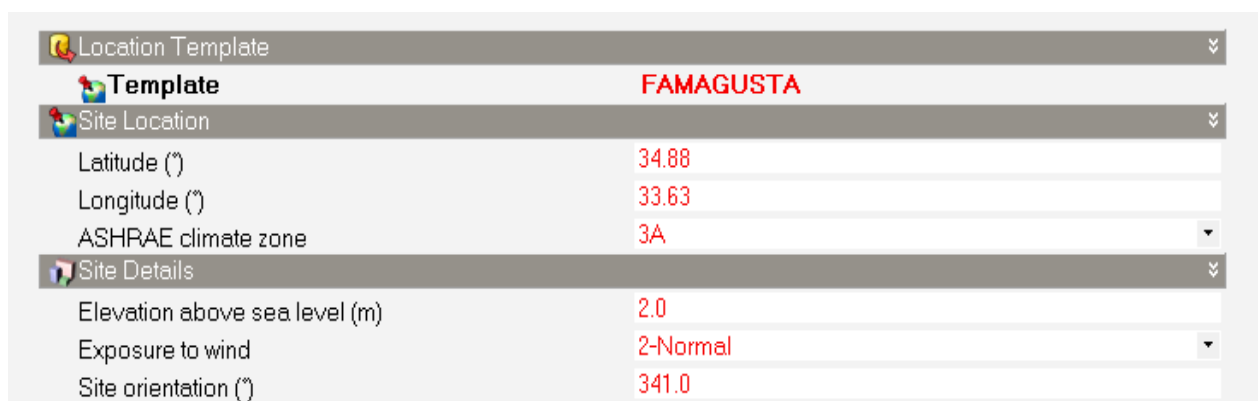


Figure 6; Famagusta location details

Basement contained 9 zones, ground floor contained 14 zones and first floor contained 14 floors as well.

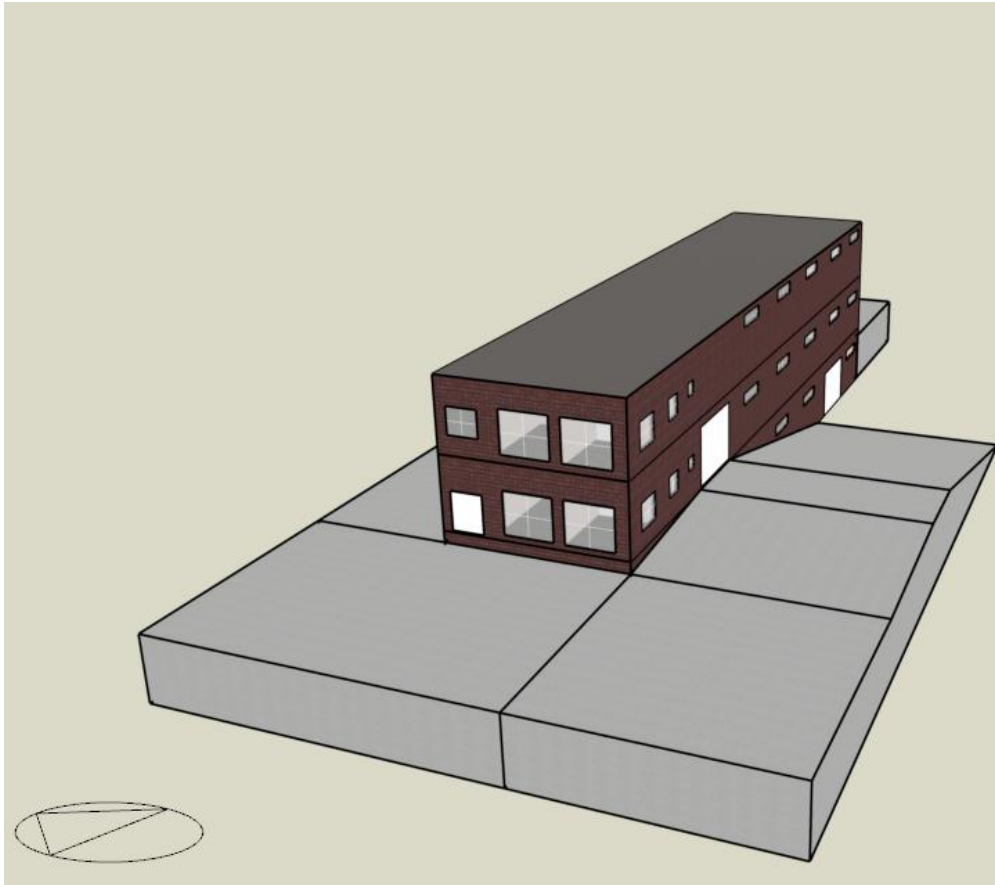


Figure 7; Roof section for PV panel placement

3. MODELLING AND SIMULATION OF PARAMETERS

The project required us to model the building in such a way that the electrical consumption per year should be around 150,000Kwh. Keeping this in mind the following parameters were adjusted.

1. Activity Template was selected as Workshop - small scale. Computers and office equipments were turned on since the building contains seven offices. The power densities were set accordingly.

Activity Template

Template Workshop - small scale

Sector B1 Offices and Workshop businesses

Zone multiplier 1

☒ Include zone in thermal calculations

☒ Include zone in Radiance daylighting calculations

Floor Areas and Volumes >>

Occupancy >>

Contaminant Generation and Removal >>

Holidays >>

DHW >>

Environmental Control >>

Computers >>

☒ On

Power density (W/m2) 3.00

Schedule Office_WkshpSS_Equip

Radiant fraction 0.200

Office Equipment >>

☒ On

Power density (W/m2) 5.00

Schedule Office_WkshpSS_Equip

Radiant fraction 0.200

Miscellaneous >>

☐ On

Catering >>

Process >>

☐ On

Figure 8; Power density allocation for workshop

- The project pdf instructions clearly stated that the frame of the building should be reinforced concrete and walls made of bricks. On design Builder external walls library, concrete, cast-reinforced was selected with a thickness of 0.25m. brickwork outer was selected with a thickness of 0.1m.

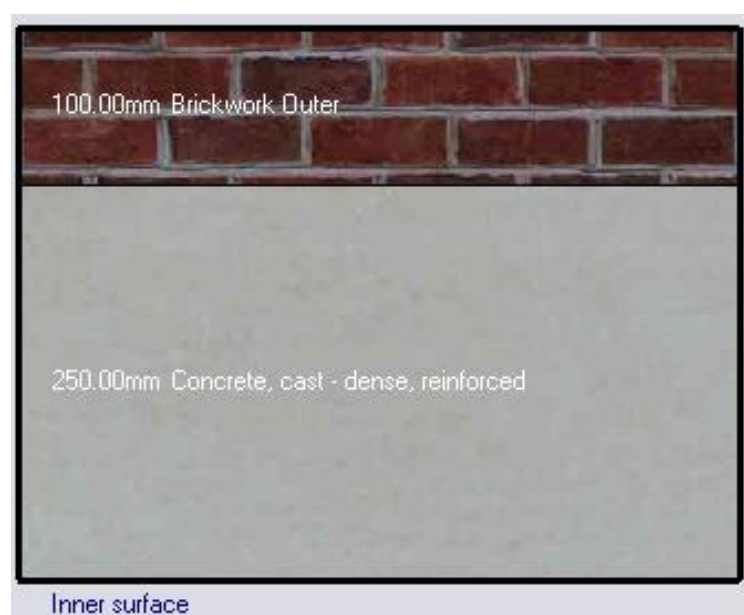


Figure 9; Wall properties of the building

3. Lighting template was selected as reference with a normalized power density of 5.
4. HVAC template was selected as split no fresh air. Mechanical ventilation was switched on. Heating and cooling also being on with electricity coming from the grid and the maximum supply temperature was set at **30** degrees and minimum supply temperature was set at **15** degrees. We then set the schedules accordingly for both heating and cooling as shown on the figures below.

Profiles							
M...	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Jan	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Feb	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Mar	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Apr	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
May	Off	Off	Off	Off	Off	Off	Off
Jun	Off	Off	Off	Off	Off	Off	Off
Jul	Off	Off	Off	Off	Off	Off	Off
Aug	Off	Off	Off	Off	Off	Off	Off
Sep	Off	Off	Off	Off	Off	Off	Off
Oct	Off	Off	Off	Off	Off	Off	Off
Nov	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Dec	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off

Figure 10; Schedule settings

Profiles							
M...	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Jan	Off	Off	Off	Off	Off	Off	Off
Feb	Off	Off	Off	Off	Off	Off	Off
Mar	Off	Off	Off	Off	Off	Off	Off
Apr	Off	Off	Off	Off	Off	Off	Off
May	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Jun	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Jul	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Aug	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Sep	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Oct	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	8:30 - 16:00	Off	Off
Nov	Off	Off	Off	Off	Off	Off	Off
Dec	Off	Off	Off	Off	Off	Off	Off

Figure 11; Schedule settings cont'd

Lastly, after carefully considering all necessary parameters, the results from the simulation showed that the electrical consumption of the building per year is 150115.65KWh per year which approximately meets the requirements of the project if rounded off to three significant figures.

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m ²]	Energy Per Conditioned Building Area [kWh/m ²]
Total Site Energy	150115.65	146.57	146.57
Net Site Energy	3600.92	3.52	3.52
Total Source Energy	327125.31	319.41	319.41
Net Source Energy	-136886.8	-133.7	-133.7

Figure 12; Building's Yearly Electricity Consumption

Given the yearly electrical consumption of the building (150,000 KWh), we can now calculate

A suitable PV brand was selected after looking at the manufacturer's catalogues. The JinkoSolar TigerNeo 66HL5-BDV (710-735Watt) Bifacial module with dual glass was chosen to be our PV panels.

66HL5-BDV 710-735 Watt

Mechanical Characteristics

Cell Type	N- type Mono-crystalline
No. of cells	132 (66×2)
Dimensions	2384×1303×33 mm
Weight	37.5 kg
Front Glass	2.0 mm, Anti-Reflection Coating
Back Glass	2.0 mm, Heat Strengthened Glass
Frame	Anodized Aluminium Alloy
Junction Box	IP68 Rated
Protection Class	Class II
IEC Fire Type	Class C
Connector Type	JK03M/JK03M2/Others*
Output Cables (Including Connector)	4.0 mm ² (+): 400 mm , (-): 200 mm or Customized Length

* MC4 and MC4-EVO2 available upon request and subject to availability

Packaging Configuration

Pallet Dimensions	1325×1121×2496 mm
Packing Detail (Two pallets = One stack)	33 pcs/pallets, 594 pcs/ 40'HQ Container

Engineering Drawings

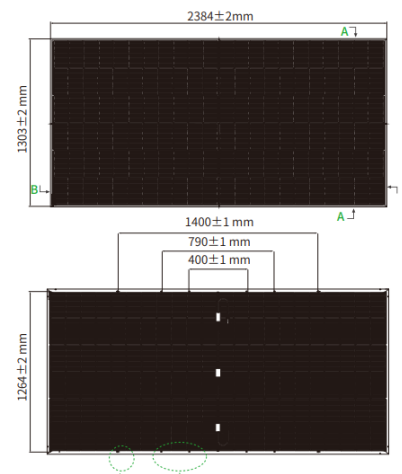


Figure 13; Properties of the PV panels from the Manufacturer's catalogue

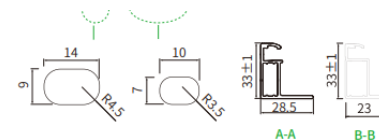
Specifications (STC)

Maximum Power - Pmax [Wp]*	710	715	720	725	730	735
Maximum Power Voltage - Vmp [V]	40.65	40.77	40.89	41.00	41.11	41.23
Maximum Power Current - Imp [A]	17.47	17.54	17.61	17.69	17.76	17.83
Open-circuit Voltage - Voc [V]	48.73	48.88	49.04	49.20	49.36	49.52
Short-circuit Current - Isc [A]	18.53	18.60	18.67	18.74	18.81	18.88
Module Efficiency STC [%]	22.86	23.02	23.18	23.34	23.50	23.66
Power Tolerance	0 ~ + 3 %					
Temperature Coefficient of Pmax	-0.29 %/°C					
Temperature Coefficient of Voc	-0.25 %/°C					
Temperature Coefficient of Isc	0.045 %/°C					

STC: Irradiance 1000W/m², Cell Temperature 25°C, AM=1.5. *Power measurement tolerance: ±3%

Specifications (BNPI)

Maximum Power - Pmax [Wp]*	784	790	795	800	805	810
Maximum Power Voltage - Vmp [V]	40.66	40.80	40.92	41.03	41.14	41.25
Maximum Power Current - Imp [A]	19.28	19.36	19.43	19.50	19.57	19.64
Open-circuit Voltage - Voc [V]	48.73	48.88	49.04	49.20	49.36	49.52
Short-circuit Current - Isc [A]	18.53	18.60	18.67	18.74	18.81	18.88



*Note: For specific dimensions and tolerance ranges, please refer to the corresponding detailed module drawings.

Electrical Performance

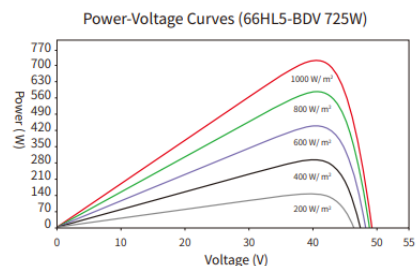


Figure 14; STC specification with max PV capacity

The panel used in this project has an efficiency of 23.66%, its maximum power rating output of 735W and a cell type N-type mono-crystalline (66 x 2). It has a weight of 37.5kg and a dimension of 2384mm x 1303mm x 33mm.

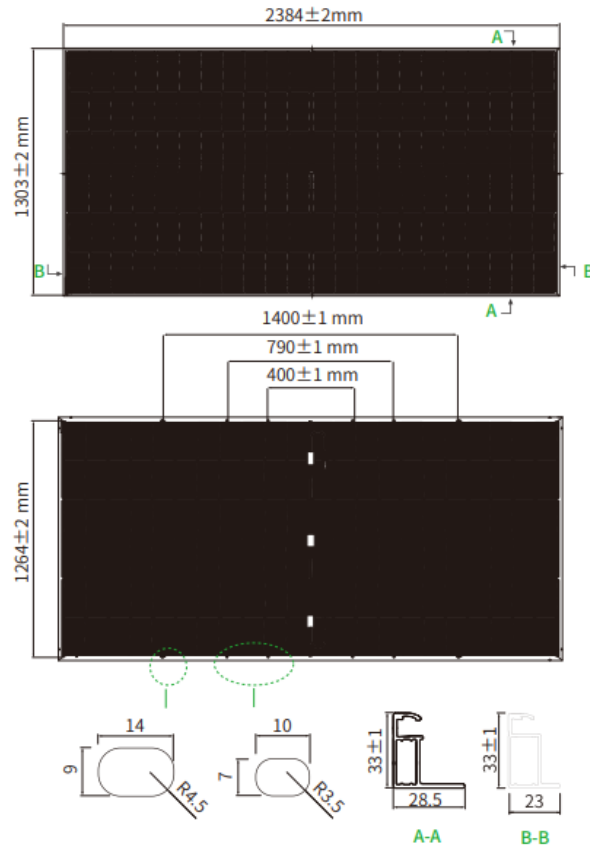


Figure 15; Engineering drawings of the PV panels used

Calculation of the area of a single panel:

$$A_p = 1.303 \times 2.384$$

$$A_p = 3.106 \text{ m}^2$$

Since the manufacturer did not provide the dimensions of each cell, it was assumed that the active area is 90% of the total panel area

$$A_{\text{active}} = 0.9 \times 3.106$$

$$A_{\text{active}} = 2.8 \text{ m}^2$$

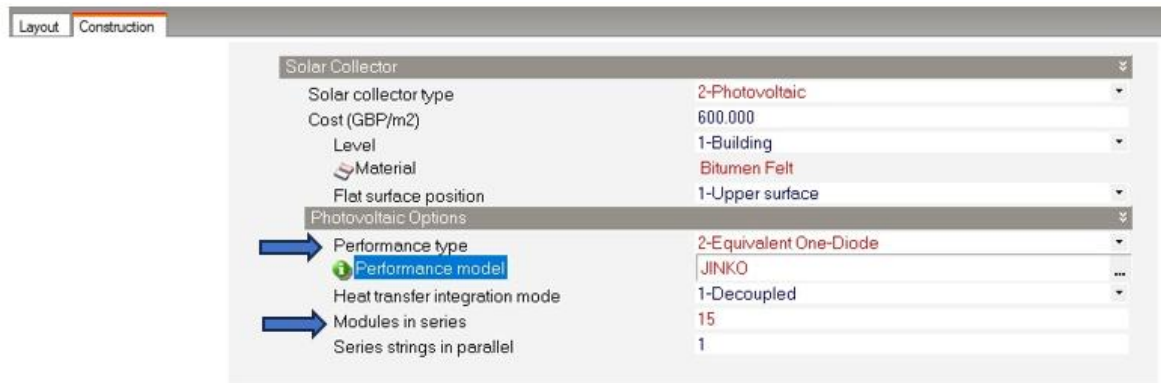


Figure 16; Performance Type

The performance type was equivalent to one-diode and our performance model was edited to be according to the manufacturer's details in the STC.

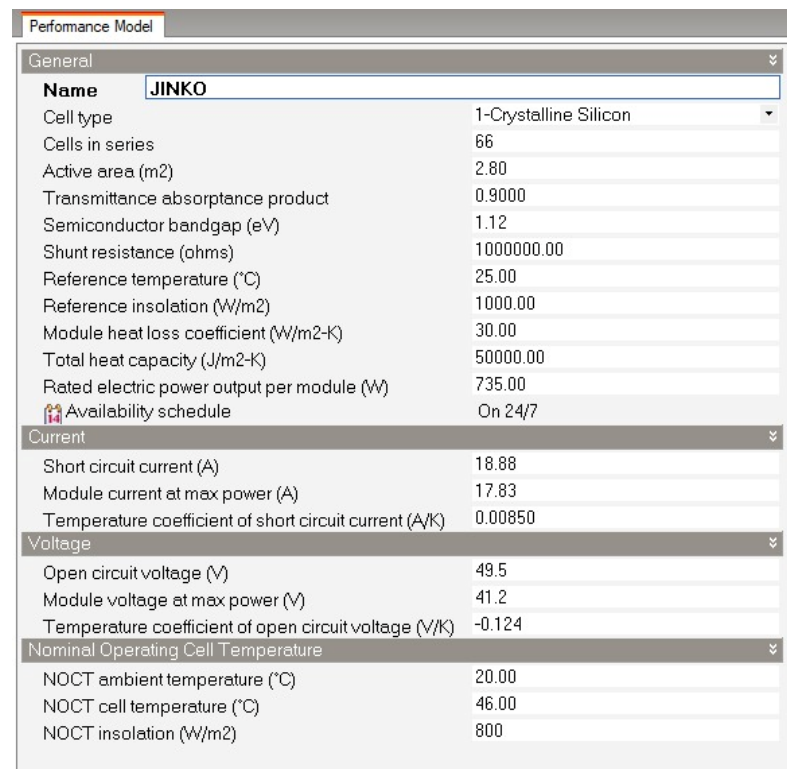


Figure 17; Edited Performance Model

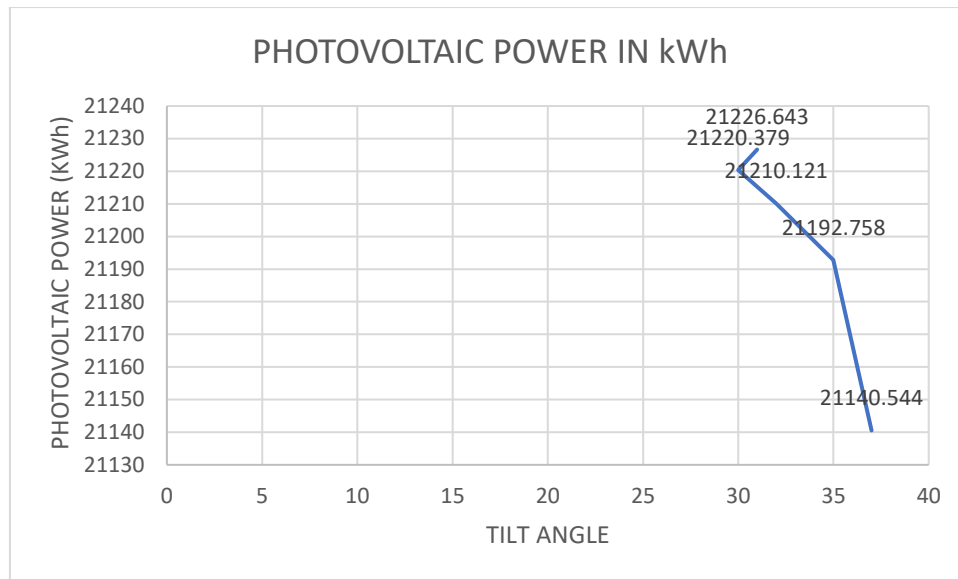
According to the manufacturer's catalogue, we had cells in series to be 66, our active area was set to be 2.8m^2 , the total rated electric power output per module was 735W. Also, the entire current and voltage sections were edited. Finally, in the nominal operating cell temperature section, the NOCT cell temperature was set to be 46°C

OBTAINING OPTIMUM TILT ANGLE

Multiple simulations were ran to obtain the appropriate optimum tilt angle. 15 panels were set to be a string and 1 string was used to run the simulation at different tilt angles. The optimum tilt angle was discovered to be 31° .

Table 1; Multiple Iterations Ran on DesignBuilder.

TILT ANGLE	PHOTOVOLTAIC POWER IN kWh
31°	21226.643
30°	21220.379
32°	21210.121
35°	21192.758
37°	21140.544



Graph 1; PV Power Against Tilt Angle

The annual electrical consumption of the building came out to 150115.65KWh

We have 15 panels connected in series (1 string) used to obtain the PV power in the above table. 7 strings were connected in parallel and the simulation was ran.

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	135673.230	174.17
Wind Power	0.000	0.00
Power Conversion	-1628.08	-2.1
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	134045.151	172.08
Electricity Coming From Utility	24511.875	31.47
Surplus Electricity Going To Utility	80658.760	103.54
Net Electricity From Utility	-56146.89	-72.1

Figure 18; Photovoltaic Power for 7 Strings

The Photovoltaic Power gotten from the 7 strings simulation was 135673.230KWh, which is not sufficient to satisfy the electrical needs of the workshop building so we re-ran the simulation with 8 strings.

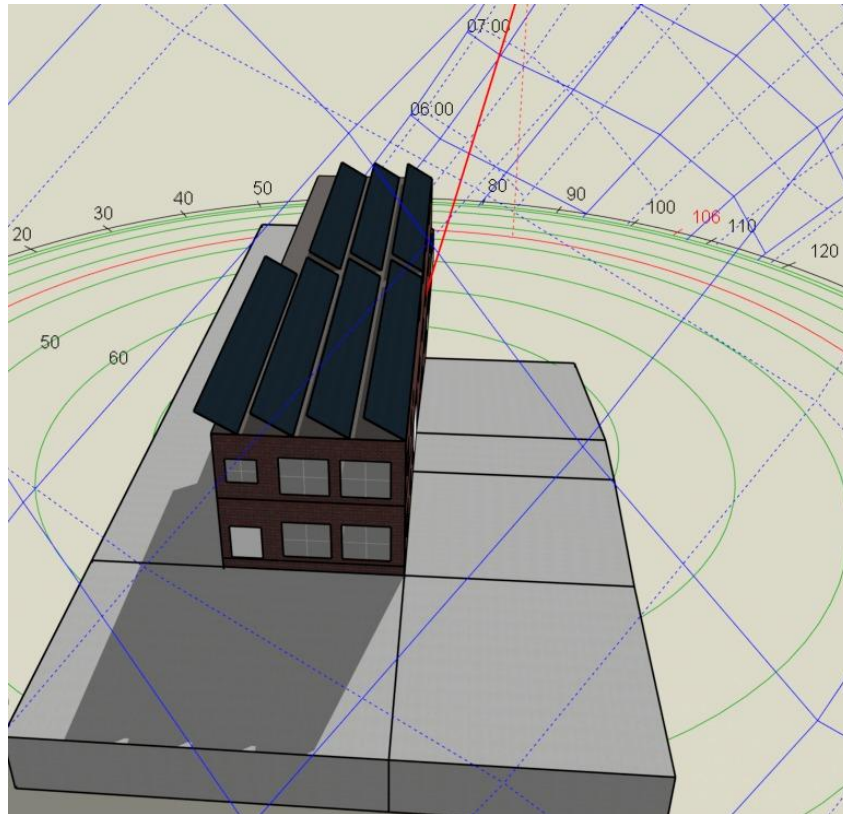


Figure 19; Panel Placement for 7 Strings

The simulation was ran again with 8 strings instead of seven strings

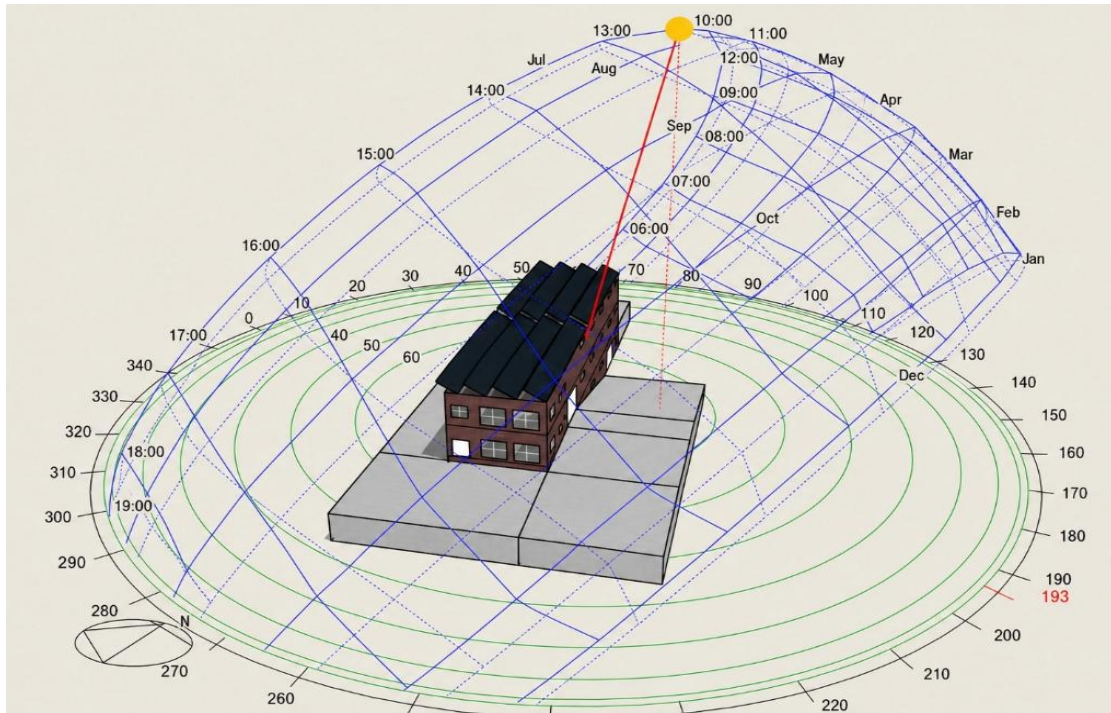


Figure 20; Panel Placement for 8 Strings

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	154226.032	197.98
Wind Power	0.000	0.00
Power Conversion	-7711.30	-9.9
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	146514.731	188.08
Electricity Coming From Utility	23564.382	30.25
Surplus Electricity Going To Utility	92180.847	118.33
Net Electricity From Utility	-68616.46	-88.1
Total On-Site and Utility Electric Sources	77898.266	100.00
Total Electricity End Uses	77898.266	100.00

Figure 21; Results from 8 Strings Simulation

Now, the PV power gotten from the 8-string simulation is sufficient for the building.

Selection of Inverters

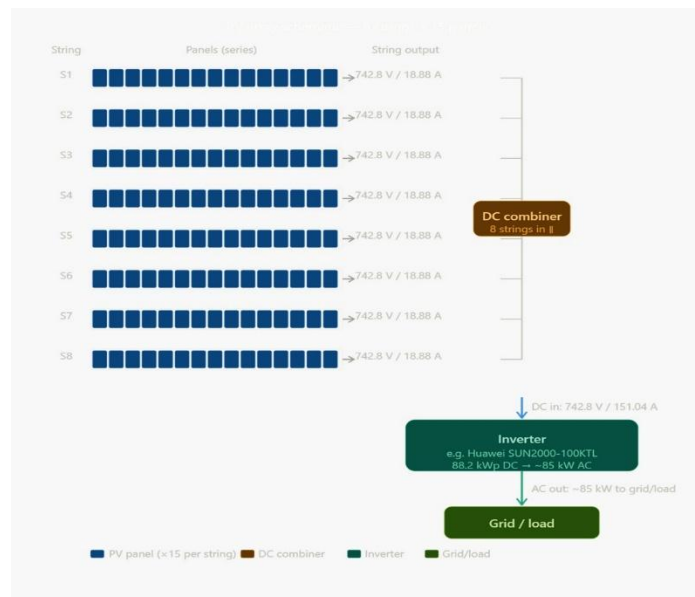


Figure 22; Schematic showing how the strings are connected

The figure above shows how the panels are connected in series and in parallel showing each string's total output voltage and current. Each panel has a V_{oc} of 49.52V, multiplying this by 15 to get the total voltage of 742.8V and the I_{sc} stays the same throughout the string as 18.88A, the current multiplies in parallel so, $18.88 \times 8 = 151.04A$.

Total V_{oc} input = 742.8V

Total I_{sc} input = 151.04A

The Huawei SUN2000-100KTL-M1 Smart PV Controller was selected as our ideal inverter type.

SUN2000-100KTL-M1
Smart PV Controller



Figure 23; Inverter type selected

SUN2000-100KTL-M1
Technical Specification

Technical Specification	SUN2000-100KTL-M1
Efficiency	
Max. Efficiency	98.8%
European Efficiency	98.6%
Input	
Max. Input Voltage	1,100 V
Max. Current per MPPT	26 A
Max. Short Circuit Current per MPPT	40 A
Start Voltage	200 V
MPPT Operating Voltage Range	200 V ~ 1,000 V
Rated Input Voltage	600 V
Number of Inputs	20
Number of MPP Trackers	10
Output	
Rated AC Active Power	100,000 W
Rated AC Apparent power	100,000 VA
Max. AC Apparent Power	110,000 VA
Max. AC Active Power (cosφ=1)	110,000 W
Rated Output Voltage	400 V, 3W + PE
Rated AC Grid Frequency	50 Hz / 60 Hz
Rated Output Current	144.4 A
Max. Output Current	160.4 A
Adjustable Power Factor Range	0.8 LG ~ 0.8 LD
Max. Total Harmonic Distortion	<3%
Protection	
Input-side Disconnection Device	Yes
Anti-islanding Protection	Yes
AC Overcurrent Protection	Yes
DC Reverse-polarity Protection	Yes
PV-array String Fault Monitoring	Yes
DC Surge Arrester	Type II
AC Surge Arrester	Type II
DC Insulation Resistance Detection	Yes
Residual Current Monitoring Unit	Yes
Communication	
Display	LED Indicators, Bluetooth/WLAN + APP
RS485	Yes
USB	Yes
MBUS	Yes (isolation transformer required)
General Data	
Dimensions (W x H x D)	1,035 x 700 x 365mm (40.7 x 27.6 x 14.4 inch)
Weight (with mounting plate)	90 kg (197.4 lb.)
Operating Temperature Range	-25°C ~ 60°C (-13°F ~ 140°F)
Cooling Method	Smart Air Cooling
Max. Operating Altitude	4,000 m (13,123 ft.)
Relative Humidity	0 ~ 100%
DC Connector	Stäubli MC4
AC Connector	Waterproof Connector + OT/DT Terminal
Protection Degree	IP66
Topology	Transformerless
Nighttime Power Consumption	≤ 3.5W
Country of Manufacture	China
Standard Compliance (more available upon request)	
Safety	EN/IEC 62109-1, EN/IEC 62109-2, IEC 62116
Grid Connection Standards	AS/NZS 4777.2:2020

Figure 24; Inverter Specifications

The MPPT operating voltage range is from 200V – 1000V and our input voltage gotten from the panels has a maximum value of 742.8V and it is well within the inverter range.

From the inverter specifications, the total number of MPP Trackers is 10 and the number of inputs is 20, which means for each MPP Trackers, two strings can be connected and we have only 8 strings, meaning only 4 maximum power point trackers is needed from the inverter.

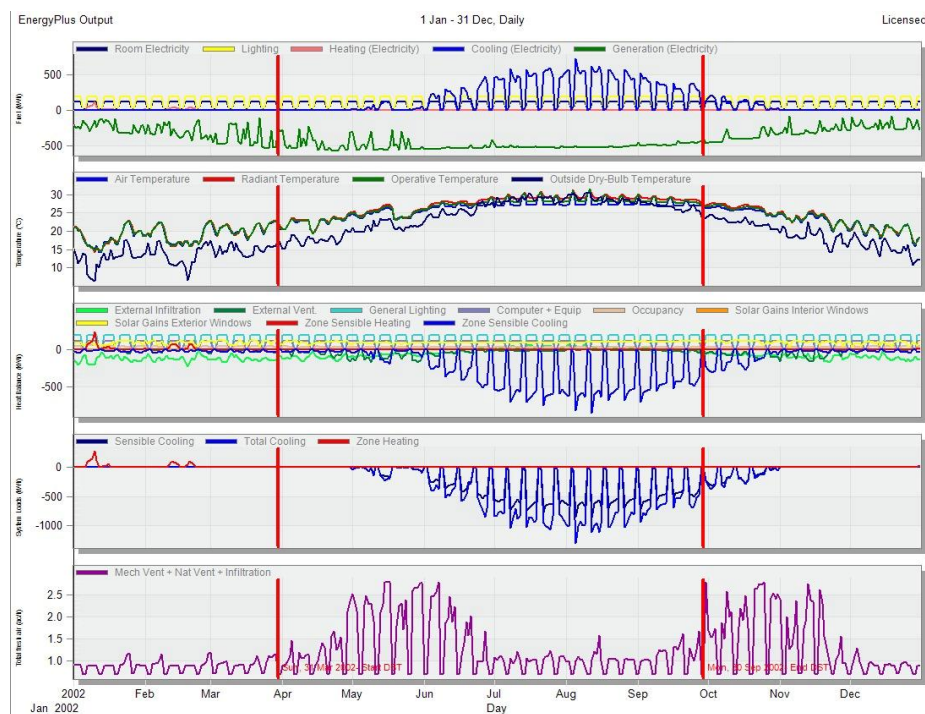
The maximum Isc per MPPT should not exceed 40A.

$$18.88\text{A} \times 2 = 37.76\text{A} < 40\text{A}$$

This signifies that our short circuit current per MPPT is within the given range. The maximum efficiency of the inverter is stated to be 98.8%.

4. RESULTS AND DISCUSSION

The annual total electrical consumption of the building calculated from design builder and it was found to be 150115.65KWh.



Graph 2; Annual Building Performance

For the annual electricity simulation, it was set to run from the 1st of January to the 31st of December which gives the above result.

The graph above shows that the building's cooling requirements peaked early august and lasted from April to the beginning of October. While the heating needs started in early October and continued up to the beginning of April. Meanwhile, the rooms' requirements for lighting and power don't change throughout the course of the year.

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	154226.032	197.98
Wind Power	0.000	0.00
Power Conversion	-7711.30	-9.9
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	146514.731	188.08
Electricity Coming From Utility	23564.382	30.25
Surplus Electricity Going To Utility	92180.847	118.33
Net Electricity From Utility	-68616.46	-88.1
Total On-Site and Utility Electric Sources	77898.266	100.00
Total Electricity End Uses	77898.266	100.00

Figure 25; Simulation ran on DesignBuilder.

From the simulation image above, our power gotten from the photovoltaic panels was 154.226Kwh which is sufficient to run the building.

$$\frac{154226.032}{150115.65} \times 100\% = 102.7\%$$

This indicates that we were able to meet 102.7% of the site's yearly power needs using the solar PV panels.

5. CONCLUSION

In this project, a photovoltaic (PV) system was successfully designed and analyzed for a three-story workshop building located in Famagusta, North Cyprus. The building was modeled using DesignBuilder software by considering realistic operational parameters such as occupancy schedules, lighting systems, HVAC operation, equipment loads, and construction materials. Through simulation, the building's annual electricity consumption was determined to be approximately 150,115.65 kWh, which closely matched the project target of 150,000 kWh per year. Using local climatic data and DesignBuilder simulations, the PV system configuration was optimized. JinkoSolar TigerNeo 66HL5-BDV bifacial modules with an efficiency of 23.66% and a power output of 735 W were selected for the design, resulting in a total of 120 photovoltaic panels connected in 8 parallel strings.

The designed PV system was further evaluated through electrical analysis and inverter selection to ensure safe and efficient operation. The Huawei SUN2000-100KTL-M1 inverter was selected due to its compatibility with the calculated voltage and current requirements of the PV array. Simulation results showed that the photovoltaic system generated approximately 154,226 kWh annually, which was capable of satisfying about 102.7% of the building's yearly electrical demand. The results also indicated that cooling loads dominated during the summer months while heating demands increased during winter periods. Overall, the project demonstrated that photovoltaic technology can effectively provide sustainable and reliable electrical energy for commercial workshop buildings while significantly reducing dependence on conventional grid electricity and supporting environmentally friendly energy production.

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APPENDIX

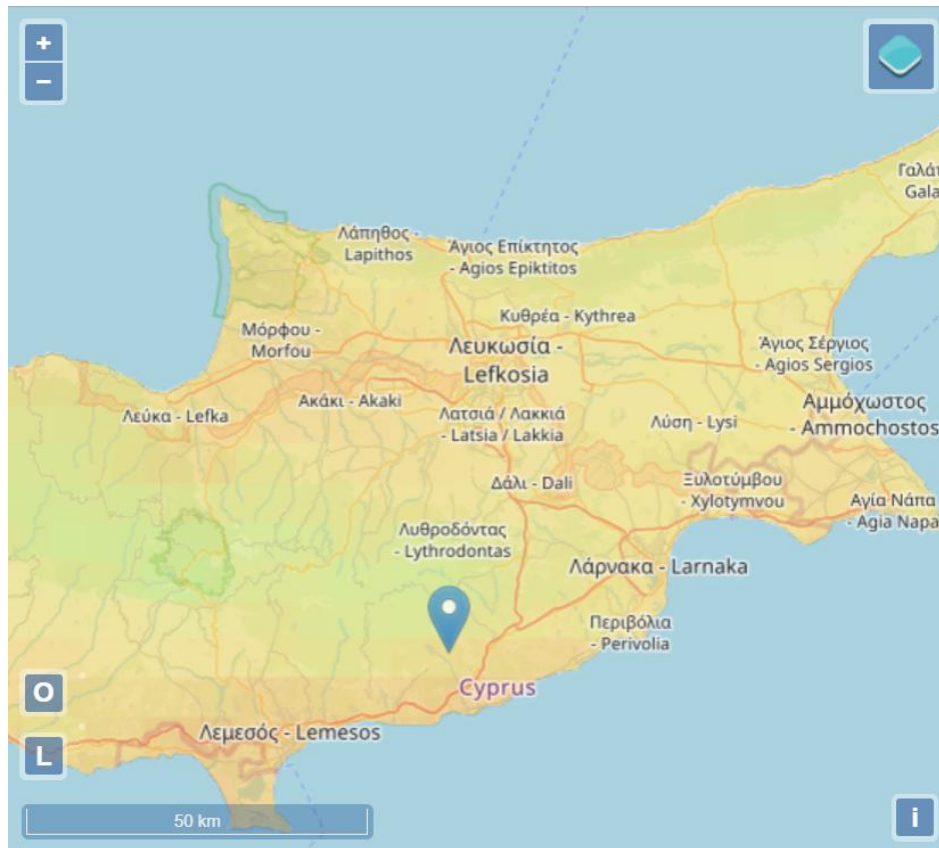


Figure 26; Location Used (Cyprus)

USE OF ARTIFICIAL INTELLIGENCE

Specific AI tools like Gemini and Claude were used in correcting and editing the grammatical errors and other minor blunders in certain sections of the report like the abstract, introduction and the conclusion. Aside from the listed sections, all other part of the report was hand crafted and calculated manually.